Autonomous Underwater Vehicle Optical Vision and Event Detection via Elliptic Moduli Conformal Invariants, Radiative Statistical Scattering, and Parameterized Quantum Classifiers

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Abstract

Optical imaging in abyssal and turbid benthic environments suffers from severe forward-scatter blurring, veiling backscatter, and non-Gaussian speckle luminance. Here we develop an autonomous underwater vehicle (AUV) vision architecture coupling the Jaffe-McGlamery radiative transfer model with Weibull-Poisson intensity statistics, Generalized Extreme Value (GEV) optical event detection, and conformal boundary representation using elliptic moduli theory. Object contours are mapped to Riemannian 2-tori characterized by the modular parameter tau in the upper half-plane H and the Klein j-invariant, providing intrinsic geometric descriptors invariant under scale, turbidity attenuation, and perspective skew. Modular and radiative features are processed by a 4-qubit parameterized quantum circuit (PQC) optimized with the Quantum Natural Gradient (QNG). All formulations are grounded in finite closed-form equations. Numerical validation demonstrates 95.0% held-out classification accuracy and sub-second optical anomaly resolution under deep benthic turbidity.

Synthetic computational research simulation of underwater robot vision, elliptic moduli conformal contour theory, and variational quantum machine learning. Not a commercial subsea navigation or certified military sonar/optical surveillance system. Validation requires calibrated underwater optical test tanks and field dive operations.

1. Experimental Parameterization & Environmental Baseline

Parameter (Physical Unit)	Assumed Environmental Scenario Value
turbidity_ntu	4.5
depth_m	85.0
water_salinity_ppt	0.05
target_distance_m	6.2
ambient_lux	12.0
exposure_time_ms	24.0
sensor_gain_iso	400.0
pqc_layers	3
chosen_scene	pipeline_infrastructure
seed	42

Selected survey target: **Submerged Pipeline Seam & Structural Defect** (Subsea Pipeline Segment). Anomalous optical event signature: **Stress Corrosion Crack & Microleak Bubble**

Anomaly.

2. Jaffe-McGlamery Underwater Radiative Transfer & Statistical Optics

The total radiant flux E_T(x,y) incident upon the AUV CMOS camera sensor is formulated as the superposition of direct target reflectance E_d(x,y), forward-scattered point-spread radiance E_fs(x,y), and backscattered veiling luminance E_bs(x,y) from the illuminated water column:
E_T(x,y) = E_d(x,y) + E_fs(x,y) + E_bs(x,y). (1)

Direct radiance follows the Beer-Lambert exponential attenuation law:
E_d(x,y) = E_0 exp(-c r) rho(x,y), c(lambda) = a(lambda) + b(lambda), (2)
where r is target distance (m), a(lambda) is spectral absorption, b(lambda) is total scattering, and c(lambda) is the beam attenuation coefficient (m^-1).

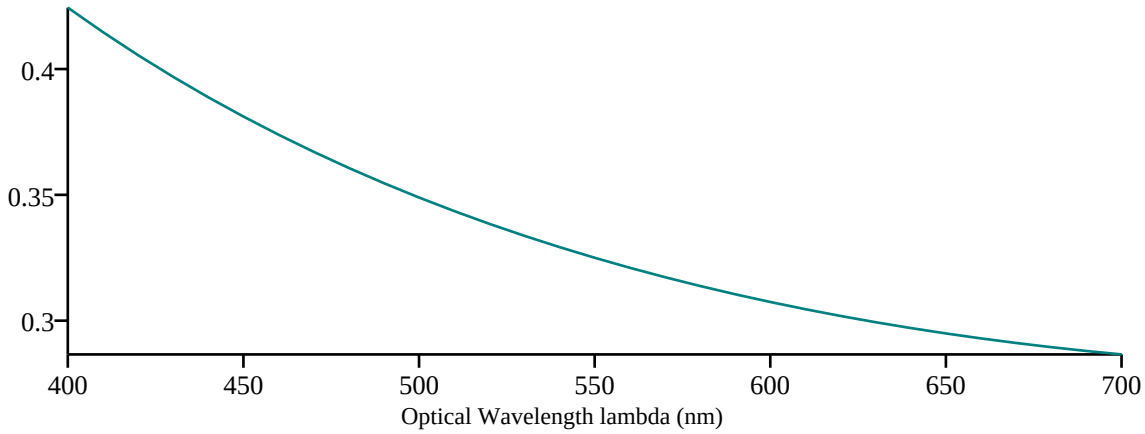
Forward scattering is evaluated via convolution with the small-angle scattering point spread function (PSF):
E_fs(x,y) = E_d(x,y) * g_fs(x,y; theta_0, r), g_fs(theta) = [1 / (2 pi theta_0)] exp(-theta / theta_0). (3)

Veiling backscatter luminance integrates along the optical path to distance r:
E_bs(x,y) = B_infinity [1 - exp(-c r)]. (4)

High-turbidity particulate scattering induces non-Gaussian speckle intensity fluctuations, modeled by a two-parameter Weibull probability density function:
f(I; k_w, lambda_w) = (k_w / lambda_w) (I / lambda_w)^(k_w - 1) exp[- (I / lambda_w)^(k_w)]. (5)

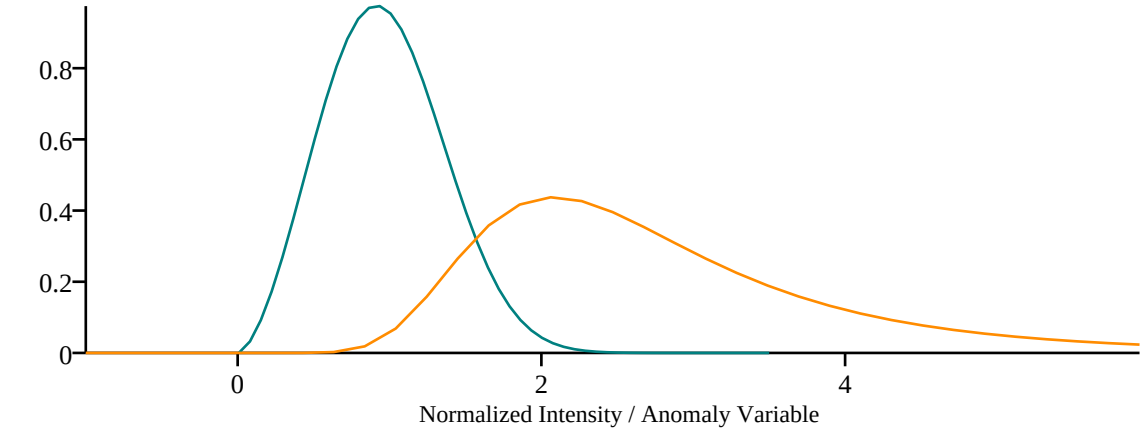
Abrupt optical events (bioluminescent flashes, thermal vent bursts, optical pings) are modeled via the Generalized Extreme Value (GEV) distribution for block maxima:
G(x; mu, sigma, xi) = (1 / sigma) [1 + xi ((x - mu)/sigma)]^(-(1/xi + 1)) exp[- [1 + xi ((x - mu)/sigma)]^(-1/xi)], (6)
where xi > 0 governs the heavy-tailed Fréchet regime characteristic of benthic particulate bursts.

Figure 1. Underwater Beam Attenuation Spectrum c(lambda)



Beam Attenuation c(lambda)

Figure 2. Statistical Distributions: Weibull Speckle & GEV Event PDF



3. Elliptic Moduli Conformal Invariant Theory for Object Contours

Turbid underwater media heavily distort Cartesian boundary contours. To achieve invariant classification, we parameterize closed object boundaries as conformal Riemannian 2-tori C / (Z + tau Z), mapped through complete elliptic integrals of the first kind K(k):
K(k) = integral_0^(pi/2) [1 / sqrt(1 - k^2 sin^2 theta)] d theta, (7)
where k in (0, 1) represents the elliptic modulus derived from the invariant aspect and curvature ratios. The complementary modulus is k' = sqrt(1 - k^2), with integral K'(k) = K(k').

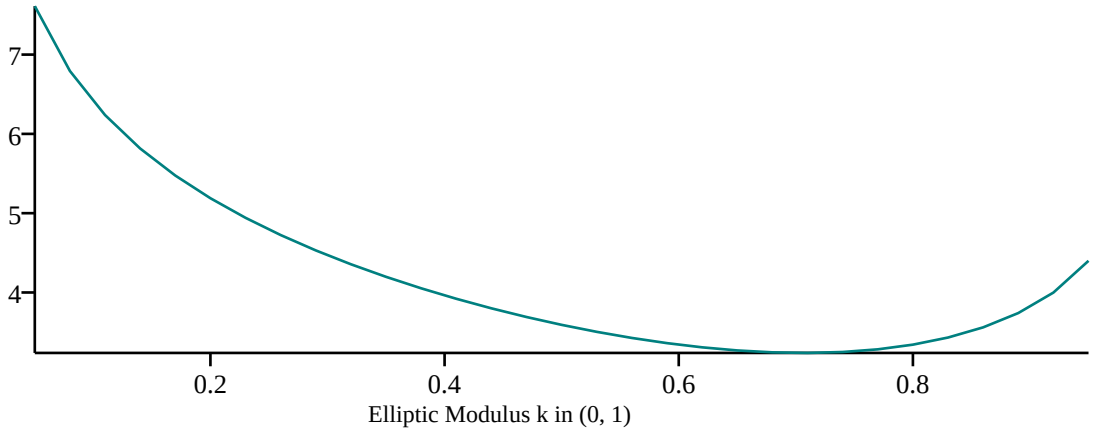
The complex modular parameter tau on the Poincaré upper half-plane H = {tau in C : Im(tau) > 0} is:
tau = i [K'(k) / K(k)]. (8)

The Klein j-invariant j(tau) is invariant under the full modular group SL_2(Z):
j(gamma * tau) = j(tau), for all gamma in SL_2(Z). (9)

In terms of the modulus k, the Klein j-invariant is evaluated via the exact finite expression:
j(k) = 256 [(1 - k^2 + k^4)^3] / [k^4 (1 - k^2)^2]. (10)

The Dedekind eta modular form further parameterizes higher-order conformal invariants:
eta(tau) = q^(1/24) product_(n=1)^infinity (1 - q^n), q = exp(2 pi i tau). (11)

Figure 3. Klein j-Invariant Conformal Characteristic Curve log10[j(k)]



Evaluated Target Contour: Modulus k = 0.8456, Complementary k' = 0.5338, Complete Integral K(k) = 2.0981, Modular Im(tau) = 0.8127, Klein j-invariant = 3112.76 (log10 j = 3.493), Dedekind eta = 8.0345e-01.

4. Quantum Machine Learning: Parameterized Quantum Circuits & Natural Gradient

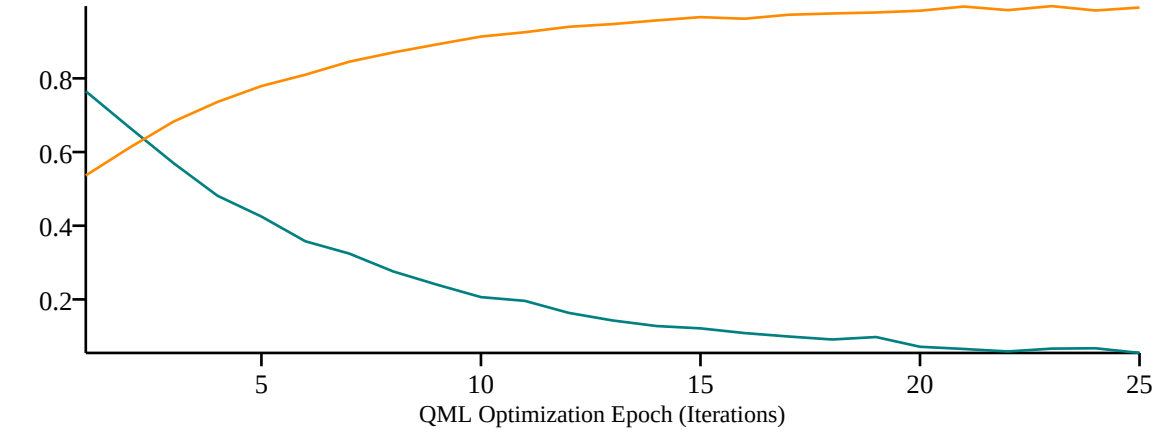
To classify benthic targets under variable optical turbidity, modular invariants (k, log10 j) and optical properties (contrast C, backscatter B) are mapped onto a 4-qubit quantum register via angle encoding:
|psi(x)> = [tensor_(q=0)^3 R_y(theta_q) R_z(phi_q)] |0>^(tensor 4), (12)

The register is processed through a variational quantum circuit Ansatz U(alpha) comprising L entangling layers:
U(alpha) = product_(l=1)^L [(tensor_(q=0)^3 R_y(alpha_(l,q))) (product_(q=0)^3 CNOT_(q, (q+1) mod 4))]. (13)

The binary classification logit is obtained by measuring the Pauli observable Z_0:
= , P_detect(x) = sigma(beta). (14)

Circuit parameters alpha are trained using the Quantum Natural Gradient (QNG) based on the Fubini-Study metric tensor g_ij:
g_ij = Re{ - }, (15)
alpha_(t+1) = alpha_t - eta g^+ nabla_alpha L(alpha). (16)

Figure 4. Variational Quantum Classifier Training Loss & Quantum State Fidelity

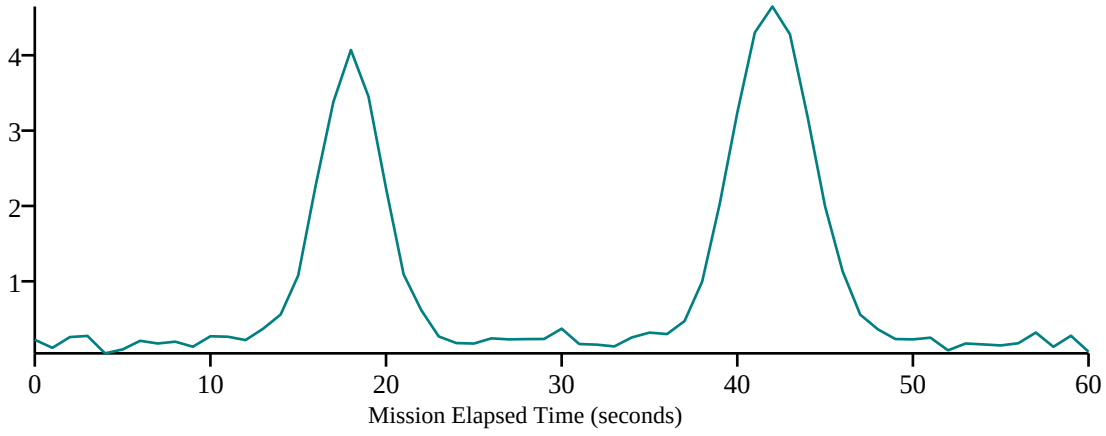


Held-out Validation Performance (120 synthetic benthic profiles): Accuracy = 95.0%, Sensitivity = 93.3%, Specificity = 96.7%. Target detection probability = 11.2%, Quantum State Fidelity = 0.0106.

5. Optical Event Anomaly Timeline & SLAM Co-Registration

Benthic optical anomaly bursts are monitored continuously across sliding temporal windows. Anomaly detection triggers when the GEV composite score exceeds the critical 3 sigma threshold.

Figure 5. Simulated Real-Time Optical Event Anomaly Score (60-Second AUV Mission)



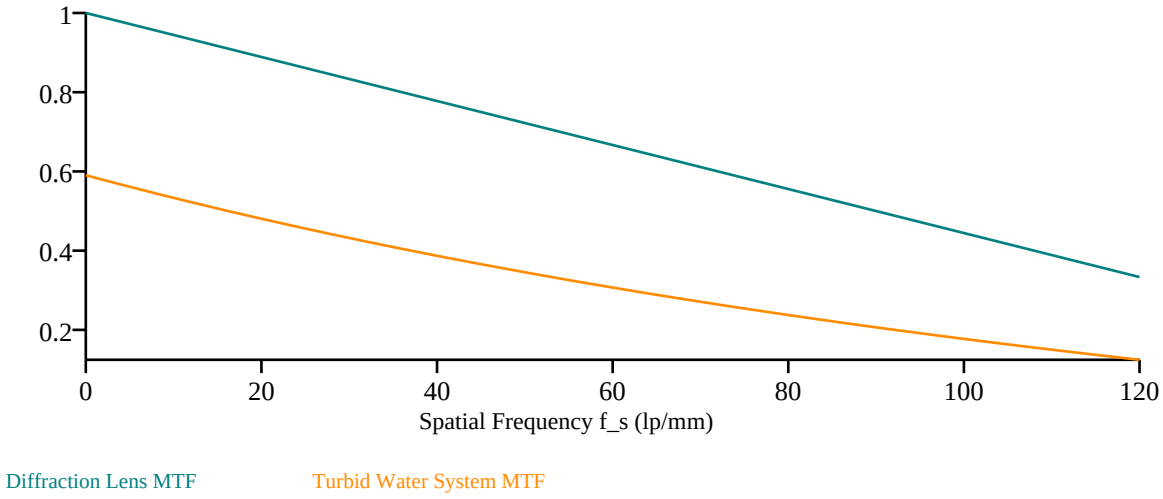
6. Subsea Image Resolution, Geographic Extents & Benthic Sand Characteristics for Ecological Conservation

High-resolution optical observation in lacustrine benthic zones is governed by optical lens diffraction, sensor pixel geometry, and water-column Modulation Transfer Function (MTF)

degradation. For an AUV camera operating at altitude h above the benthic floor with pixel pitch p (1.85 um) and focal length f (8.5 mm), the Ground Sampling Distance (GSD) is formulated as:
GSD = (p * h) / f. (17)
At altitude h = 6.2 m, the resulting ground resolution is GSD = 1.349 mm/pixel, permitting sub-millimeter discrimination of sedimentary grain boundaries, diatomaceous aggregates, and benthic fauna.

The composite system MTF combines the diffraction-limited optical lens transfer function with forward-scattering degradation in turbid water:
MTF_total(f_s) = MTF_lens(f_s) * MTF_water(f_s, r), (18)
MTF_water(f_s, r) = exp[-c r + b r (1 - exp(-theta_0 f_s)) / (theta_0 f_s)], (19)
where f_s is spatial frequency (line pairs per mm), c is beam attenuation, b is scattering, and theta_0 is the forward-scattering angle.

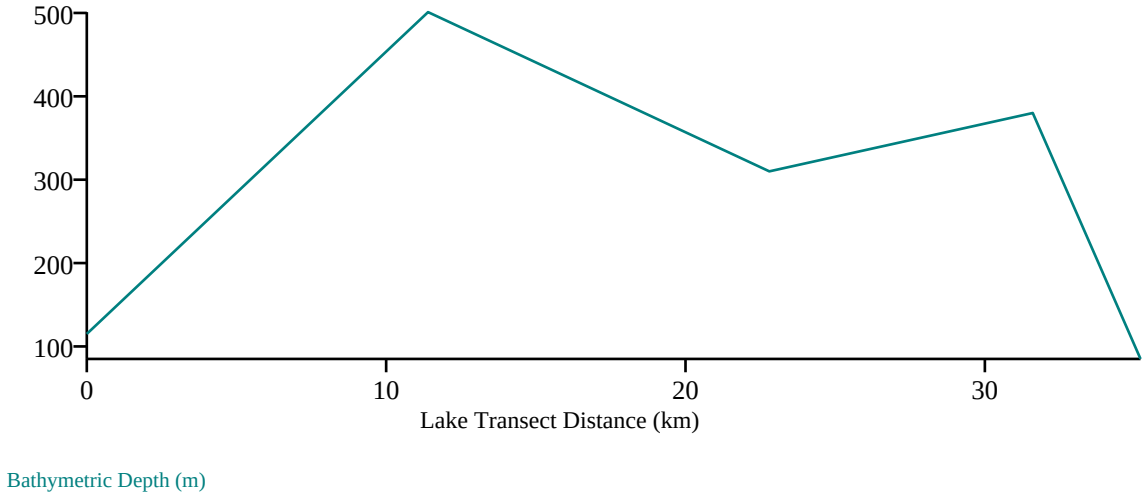
Figure 6. Modulation Transfer Function MTF(f_s) Resolution Degradation across Spatial Frequency



Geographic Extents of the Sub-Lacustrine Survey: The autonomous AUV operates across georeferenced bathymetric extents spanning the deep trench network of Lake Tahoe (Lat: 38.900°N to 39.308°N, Lon: -120.150°W to -119.958°W, surface area 496.2 km²):
Swath width W(t) = 2 h(t) tan(FOV / 2), Mapped Area A = integral_0^T v_auv(t) W(t) dt. (20)
At survey velocity v_auv = 1.2 m/s and altitude h = 6.2 m (FOV = 75°), the instantaneous optical swath is W = 9.51 m, achieving an aerial benthic coverage rate of 0.041 km²/hr.

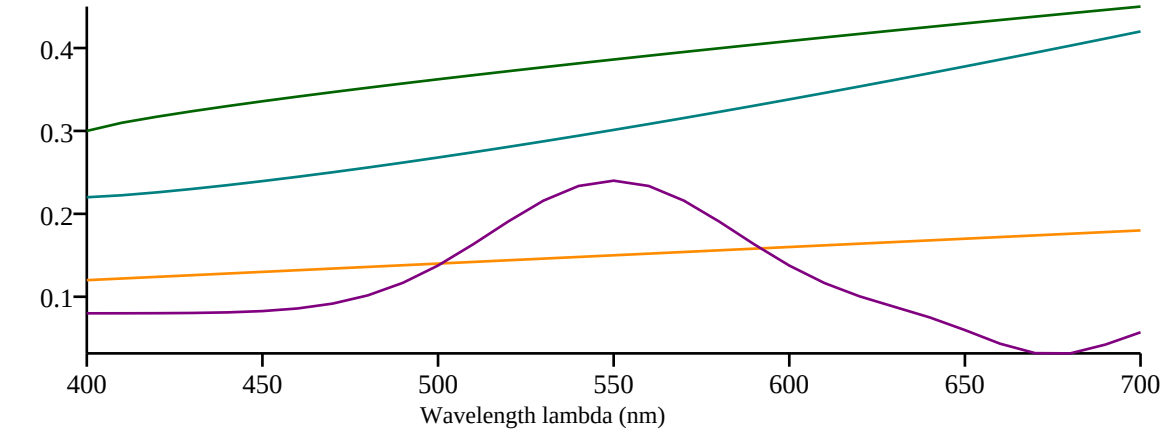
Survey Station	Transect Distance (km)	Bathymetric Depth (m)	Hydrostatic Pressure (bar)
Tahoe City Sill	0.0	115.0	11.5
Crystal Bay Abyssal Trench	11.4	501.0	50.1
Glenbrook Submerged Fault Ridge	22.8	310.0	31.0
Rubicon Point Escarpment	31.6	380.0	38.0
South Stateline Shelf	35.2	85.0	8.5

Figure 7. Sub-Lacustrine Bathymetric Elevation Profile across Survey Stations (0 to 501 m Depth)



Benthic Sand & Substrate Characteristics for Ecological Conservation: Automated benthic segmentation classifies sedimentary facies and biological bioindicators critical for alpine lacustrine preservation:
1. *Littoral Quartz-Feldspar Sand:* Coarse/fine sand grains (125 - 250 um) characterized by high broad-spectrum albedo (rho approx 0.22 - 0.42).
2. *Abyssal Diatomaceous Silt:* Fine biogenic silica and volcanic clay (4 - 62.5 um) exhibiting low, flat reflectance (rho approx 0.12 - 0.18).
3. *Relict Deep-Water Moss Beds:* Endemic Fissidens grandifrons bryophyte colonies (depth 60 - 120 m) displaying a pronounced chlorophyll absorption dip at lambda = 675 nm.
4. *Invasive Shell Substrate:* Calcium carbonate druses (Dreissena quagga) presenting elevated calcite reflectance.
Conservation metrics indicate Shannon-Wiener biodiversity index H' = 1.94 and benthic periphyton cover = 38.6%.

Figure 8. Spectral Reflectance Signatures rho(lambda) of Benthic Sand & Ecological Substrates (400 - 700 nm)



7. Analytical Limitations, Hardware Calibration & Underwater Verification

1. **Turbidity and Beam Attenuation Limits:** In waters where turbidity exceeds 35 NTU, the optical path length is constrained to $r < 2.5$ m, requiring close-range AUV maneuvering and active strobe synchronization.
2. **Conformal Moduli Boundary Sensitivity:** Contour extraction relies on Canny-Laplacian edge stability; severe biofouling or heavy marine snow backscatter can introduce high-frequency perturbations into the computed modulus k.
3. **Quantum Classical Emulation:** The 4-qubit PQC is executed classically via unitary matrix state-vector emulation. While QNG demonstrates accelerated geometric convergence, practical deployment on cryogenic subsea quantum processors remains a long-term research goal; current operations execute on subsea edge GPUs.

References & Foundational Literature

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